Calm Before the Storm: Detecting Deviations from Poisson Timing in Simulated Seizure Dynamics

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1 Introduction

Epileptic seizures are often modelled as events that occur "at random," meaning the timing of seizure-related neural events is represented using a Poisson process. Under this assumption, inter-event intervals follow an exponential distribution with rate λ, whose defining property is memorylessness:

$$P(T > s + t \mid T > s) = P(T > t).$$

If this holds, past intervals provide no information about future ones.

However, clinical observations suggest otherwise. Many individuals with epilepsy report *auras*—brief sensory or emotional disturbances that precede seizures. Dynamical systems theory offers a parallel: as systems approach major transitions, variability often decreases and dynamics become more regular (*critical slowing down*).

If the brain undergoes such a transition, it should leave a statistical footprint in the inter-event intervals themselves, independent of EEG amplitude or other high-dimensional features. This leads to the central question:

> *Do inter-event intervals remain exponential as a seizure approaches, or do they show measurable deviations from Poisson-like randomness?*

2 Methods

2.1 Simulated Timing Regimes

To isolate changes in variability, I simulate $N = 5000$ intervals under three regimes:

Poisson baseline. Intervals are i.i.d. exponential,

$$f(t \mid \lambda) = \lambda e^{-\lambda t},$$

representing fully random timing.

Bursting regime. The process alternates between a baseline rate λ_0 and a higher rate λ_{burst}, with intervals remaining exponential in each phase. This tests whether increased rate alone can mimic pre-seizure structure.

Pre-seizure narrowing. The first half matches the Poisson baseline; the second half follows Gamma($k > 1$, θ):

$$f(t \mid k, \theta) = \frac{1}{\Gamma(k)\theta^k} t^{k-1} e^{-t/\theta}.$$

For $k > 1$ intervals cluster more tightly around the mean, modelling reduced variability prior to seizure onset.

2.2 Fitting the Exponential Model

Each sliding window of intervals $\{\Delta t_i\}$ is fit with the maximum-likelihood exponential rate:

$$\hat{\lambda} = \frac{1}{\overline{\Delta t}}.$$

2.3 Entropy Gap

Empirical entropy from histogram probabilities $\{p_i\}$ is

$$H_{\text{emp}} = -\sum_i p_i \log p_i.$$

An exponential($\hat{\lambda}$) has entropy

$$H_{\text{exp}} = 1 - \log(\hat{\lambda}).$$

The *entropy gap*

$$G = H_{\text{exp}} - H_{\text{emp}},$$

increases when empirical intervals become more concentrated (less random) than the exponential model.

2.4 KL Divergence

To quantify full distributional mismatch between empirical P and fitted exponential Q:

$$D_{\text{KL}}(P \,\|\, Q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

2.5 Likelihood Ratio

Let $\log L_{\text{exp}}$ and $\log L_{\Gamma}$ be log-likelihoods of the exponential and Gamma fits. The statistic

$$\text{LLR} = 2(\log L_{\Gamma} - \log L_{\text{exp}})$$

is positive when the Gamma (reduced-variability) model explains the data better.

3 Results

The three regimes produce distinct patterns across metrics. Figure 1 shows the four most diagnostic outputs: Poisson and pre-seizure histograms, KL divergence, and LLR.

Poisson baseline. The histogram matches the exponential curve; KL divergence remains near zero; LLR stays centred around zero. This confirms the method does not falsely detect structure in random data.

Bursting regime. Short-interval bursts appear, but randomness remains exponential within each phase. Entropy and KL fluctuate briefly yet return to baseline, and LLR shows no preference for the Gamma model. Increased rate alone does not resemble a seizure precursor.

Pre-seizure narrowing. Once the Gamma($k > 1$) phase begins, variability decreases. This yields: (1) mismatch between histogram and exponential fit; (2) a sustained rise in KL divergence; (3) a strong positive LLR. Together these indicate a genuine structural change, distinct from simple rate increases.

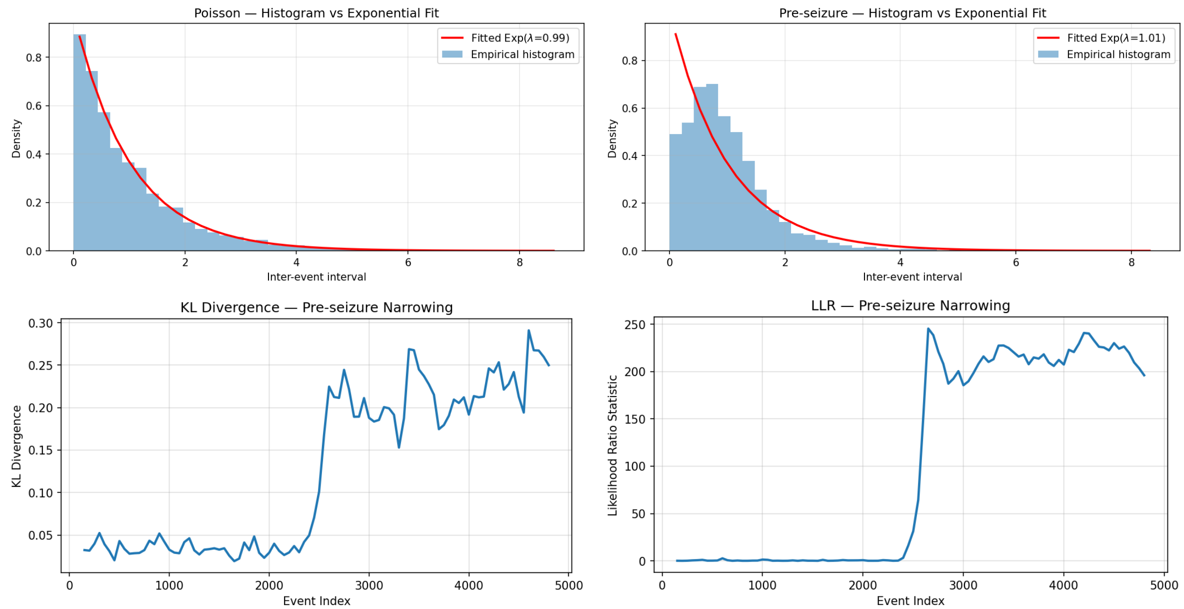

Figure 1: Top: Poisson vs. pre-seizure histograms and exponential fits. Bottom: KL divergence and LLR during pre-seizure narrowing.

4 Conclusion

Across all simulations, only the reduced-variability regime produces clear, stable deviations from exponential timing. Faster firing (bursting) does not mimic the structure seen in pre-seizure narrowing.

This suggests that *randomness itself* carries meaningful physiological information. If the brain enters a more constrained dynamical state before a seizure, this transition appears directly in the timing of neural events. A loss of variability acts as a clean, interpretable probabilistic marker of seizure-like dynamics. Interval-based analysis therefore offers a simple, transparent complement to feature-heavy approaches for studying seizure onset. Although this project is not a seizure-prediction system, the results suggest that loss of temporal randomness may begin well before overt clinical auras appear. In real neural data, such changes - if they exist - could offer a lightweight and interpretable signal for forecasting seizure risk. Timing-based metrics may therefore complement existing EEG-based prediction methods by detecting early shifts in neural variability before more noticeable symptoms emerge.

APPENDIX

A Use of AI Tools

Throughout the development of this project, I used a number of AI tools to support specific parts of the workflow while maintaining full authorship and control over the statistical design, analysis, and interpretation.

ChatGPT was used primarily as a writing and explanation assistant. I relied on it to help refine mathematical descriptions (e.g., the Gamma distribution, the intuition behind entropy, the role of KL divergence), identify redundancies in earlier drafts, and tighten the overall narrative structure.

For the video component, I used **NotebookLM**, which helped me brainstorm narrative flow, tone, and pedagogical framing. NotebookLM was particularly useful for ideation, generating alternative ways to introduce memorylessness, constructing analogies for reduced randomness (such as the bus arrival example), and shaping the pacing of the explainer-style script.

B Exponential MLE Derivation

Let $\Delta t_1, \ldots, \Delta t_m$ be inter-event intervals in a window, assumed i.i.d. exponential with rate λ. The likelihood is

$$L(\lambda) = \prod_{i=1}^{m} \lambda e^{-\lambda \Delta t_i} = \lambda^m \exp\Big(-\lambda \sum_{i=1}^{m} \Delta t_i \Big).$$

Taking logs,

$$\log L(\lambda) = m \log \lambda - \lambda \sum_{i=1}^{m} \Delta t_i.$$

Differentiate with respect to λ and set to zero:

$$\frac{d}{d\lambda} \log L(\lambda) = \frac{m}{\lambda} - \sum_{i=1}^{m} \Delta t_i = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{m}{\sum_i \Delta t_i} = \frac{1}{\overline{\Delta t}}.$$

The second derivative,

$$\frac{d^2}{d\lambda^2} \log L(\lambda) = -\frac{m}{\lambda^2} < 0,$$

confirms that this solution is a maximum. Intuitively, the best exponential fit simply uses the reciprocal of the sample mean waiting time.

C Entropy and Histogram Approximation

For a continuous distribution, Shannon entropy is

$$H = -\int f(t) \log f(t)\, dt.$$

For an exponential density $f(t \mid \lambda) = \lambda e^{-\lambda t}$, a standard calculation yields

$$H_{\exp}(\lambda) = 1 - \log \lambda.$$

For empirical data, we approximate entropy using a histogram. Suppose we partition the interval axis into bins $B_1, \dots, B_K$ and let p_i be the fraction of observations that fall in bin B_i. These $\{p_i\}$ form a discrete distribution, and we compute

$$H_{\text{emp}} = -\sum_{i=1}^{K} p_i \log p_i.$$

If the data came exactly from an exponential(λ) distribution and we used infinitely fine bins, then H_{emp} would converge to $H_{\text{exp}}(\lambda)$. In practice there is sampling noise and histogram error, but comparing H_{emp} with $H_{\text{exp}}(\hat{\lambda})$ still indicates whether the empirical intervals are more or less spread out than a fitted exponential with the same mean.

The *entropy gap*

$$G = H_{\text{exp}}(\hat{\lambda}) - H_{\text{emp}}$$

therefore has a direct interpretation:

- $G \approx 0$: intervals are as random as an exponential;
- $G > 0$: empirical distribution has lower entropy (more concentration, more regularity);
- $G < 0$: empirical distribution is more diffuse or multimodal than exponential.

D KL Divergence

Kullback–Leibler divergence measures how different two distributions are in terms of the information needed to code samples from one when using a code optimized for the other. For discrete distributions $P = \{p_i\}$ and $Q = \{q_i\}$ over the same bins,

$$D_{\text{KL}}(P \,\|\, Q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

In this project, P is the empirical histogram of inter-event intervals and Q is the histogram of the fitted exponential model. If P and Q coincide, then $D_{\text{KL}} = 0$. As P drifts away from Q (for example, by becoming more peaked), D_{KL} increases. This gives a more detailed view than entropy alone, because it is sensitive to where the probability mass moves, not just to overall spread.

E Likelihood Ratio: Gamma vs Exponential

The Gamma family with shape k and scale θ has density

$$f_\Gamma(t \mid k, \theta) = \frac{1}{\Gamma(k)\theta^k} t^{k-1} e^{-t/\theta}.$$

When $k = 1$, this reduces to the exponential density $f_{\text{exp}}(t \mid \lambda)$ with $\lambda = 1/\theta$. Thus the exponential model is a special case (a *nested* model) inside the Gamma family.

Given a sample $\{\Delta t_i\}$, I estimate $(\hat{k}, \hat{\theta})$ for the Gamma model (numerically) and $\hat{\lambda} = 1/\overline{\Delta t}$ for the exponential. The log-likelihoods are

$$\log L_{\text{exp}} = \sum_i \log f_{\text{exp}}(\Delta t_i \mid \hat{\lambda}), \qquad \log L_\Gamma = \sum_i \log f_\Gamma(\Delta t_i \mid \hat{k}, \hat{\theta}).$$

The likelihood ratio statistic is

$$\mathrm{LLR} = 2(\log L_\Gamma - \log L_{\mathrm{exp}}).$$

Intuitively, $\log L_{\mathrm{exp}}$ answers: "How surprised is an exponential model by these data?", and $\log L_\Gamma$ answers the same question for a more flexible Gamma model. If the data are truly exponential, the Gamma model cannot improve the likelihood much, so LLR stays near zero. If intervals are more regular than exponential (for example, in the pre-seizure narrowing regime), the Gamma model can choose $k > 1$ and assign much higher probability to the observed sample, producing large positive LLR.

In classical statistics one can compare LLR to a chi-square distribution to perform a formal hypothesis test, but for this project it is sufficient to interpret LLR as a continuous measure of "how strongly the data prefer a less-random model."

F Additional Figures and Regime-Specific Behavior

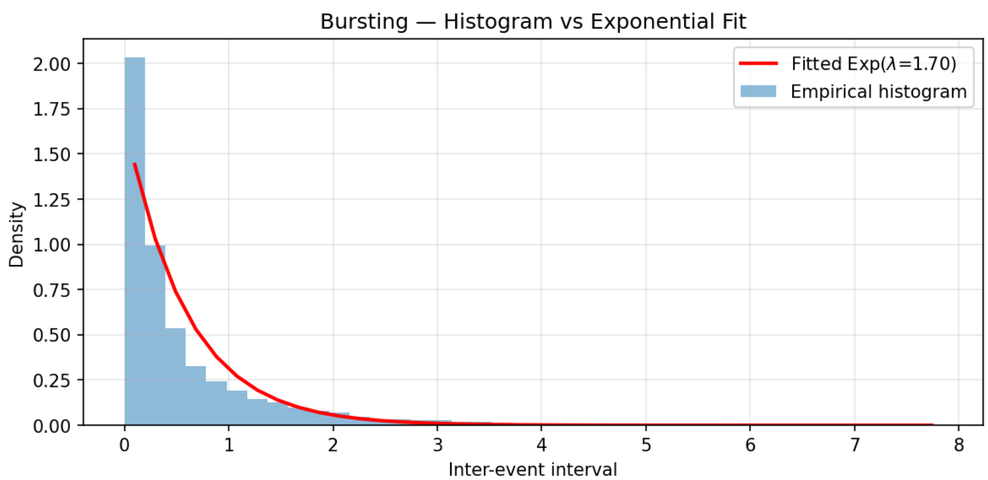

Figure 2: Bursting regime: histogram vs fitted exponential. Short intervals are common during bursts, but the overall shape is still broadly exponential.

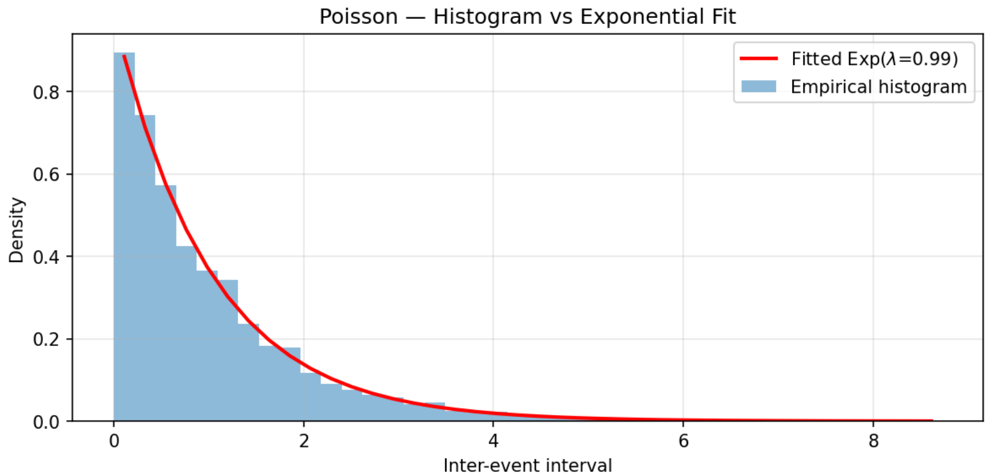

Figure 3: Poisson baseline: excellent agreement with the exponential model.

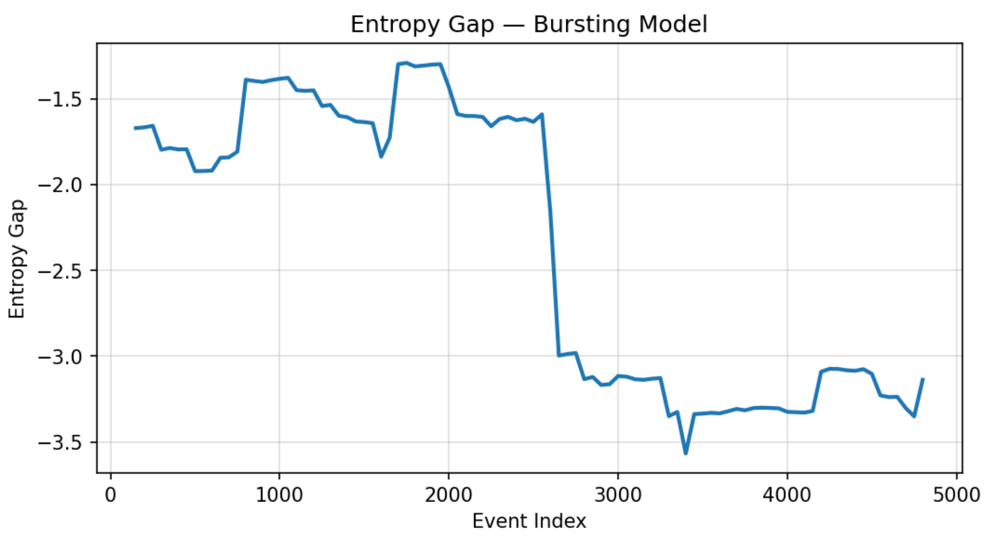

Figure 4: Entropy gap for the bursting regime. Fluctuations occur when bursts change the local rate, but there is no sustained shift away from zero.

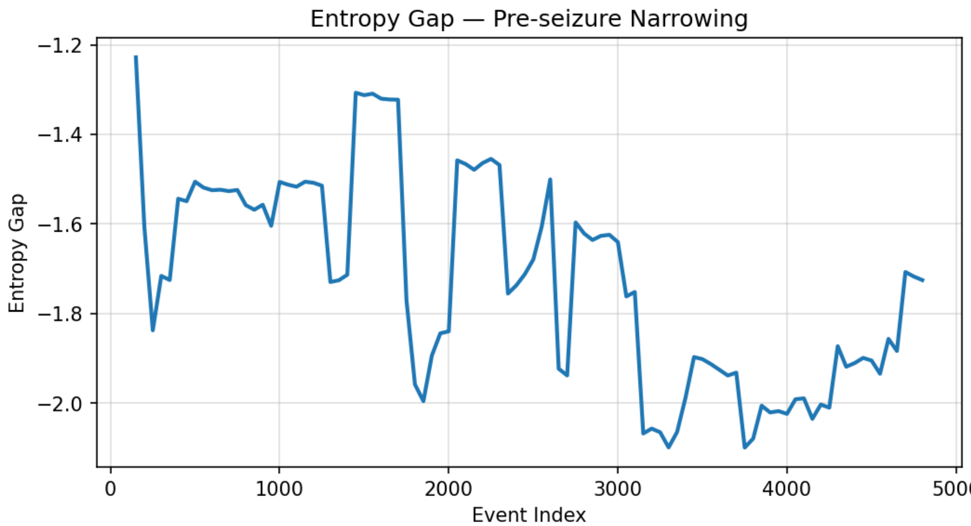

Figure 5: Entropy gap for the pre-seizure narrowing regime. After the transition, G becomes positive and remains elevated, indicating persistent loss of randomness.

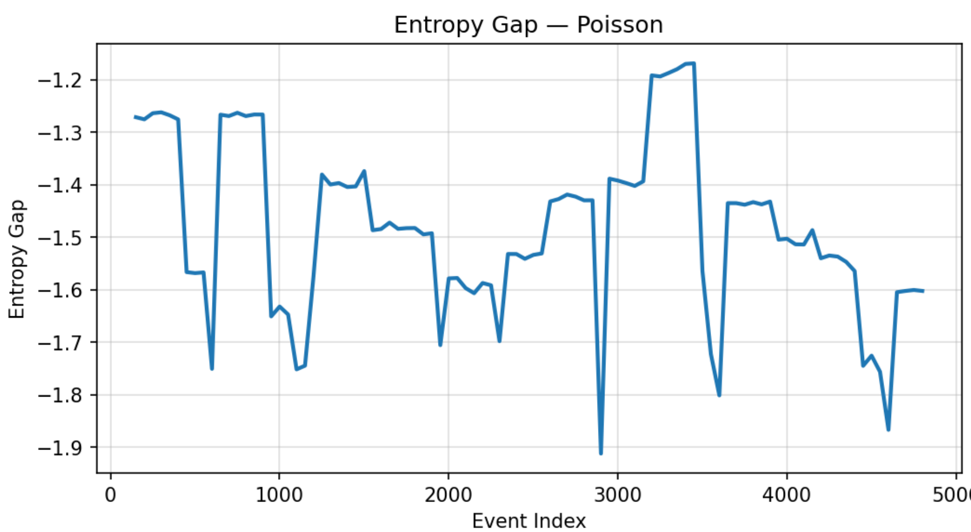

Figure 6: Entropy gap for the Poisson baseline. Values fluctuate around zero, as expected for a truly exponential process.

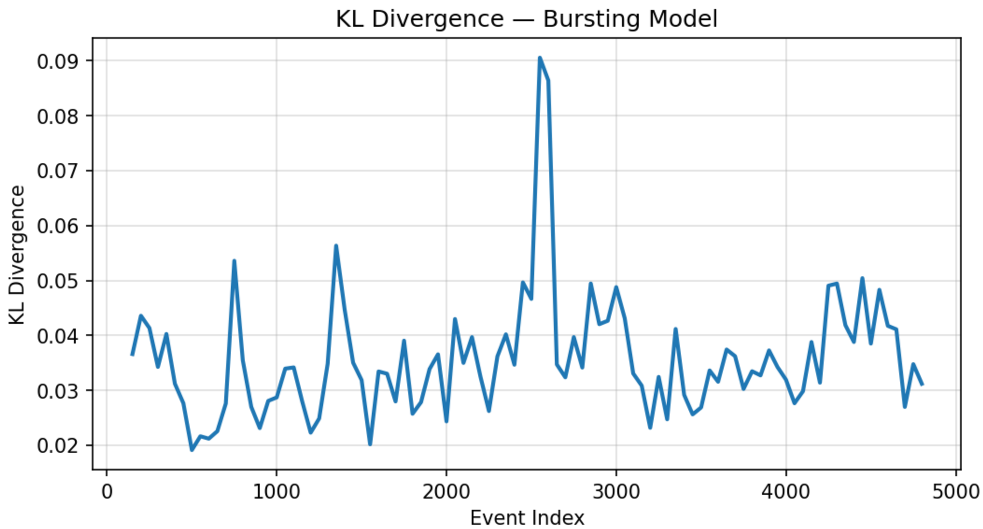

Figure 7: KL divergence for the bursting regime. Divergence spikes modestly during bursts but returns toward baseline afterwards.

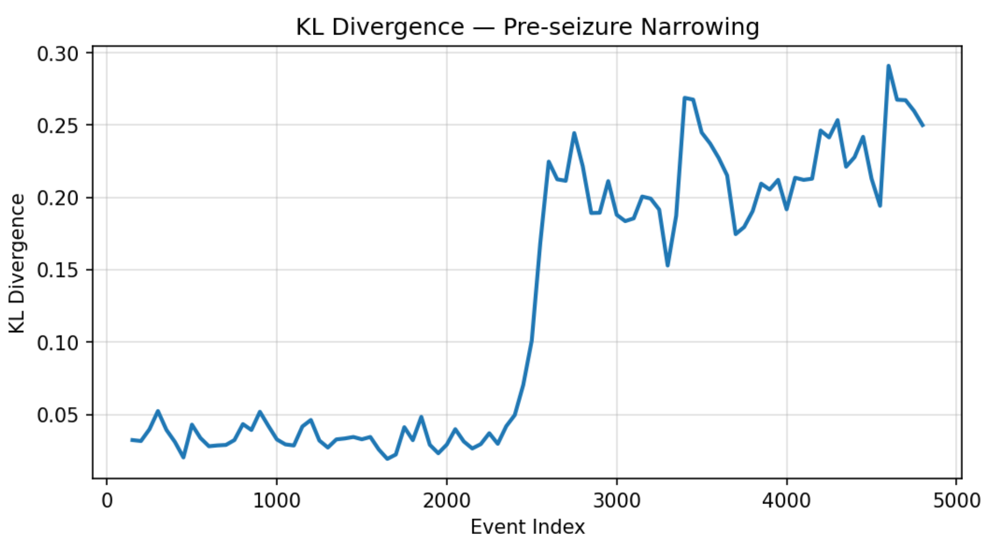

Figure 8: KL divergence for the pre-seizure narrowing regime. A clear step up signals that the exponential model no longer matches the empirical distribution.

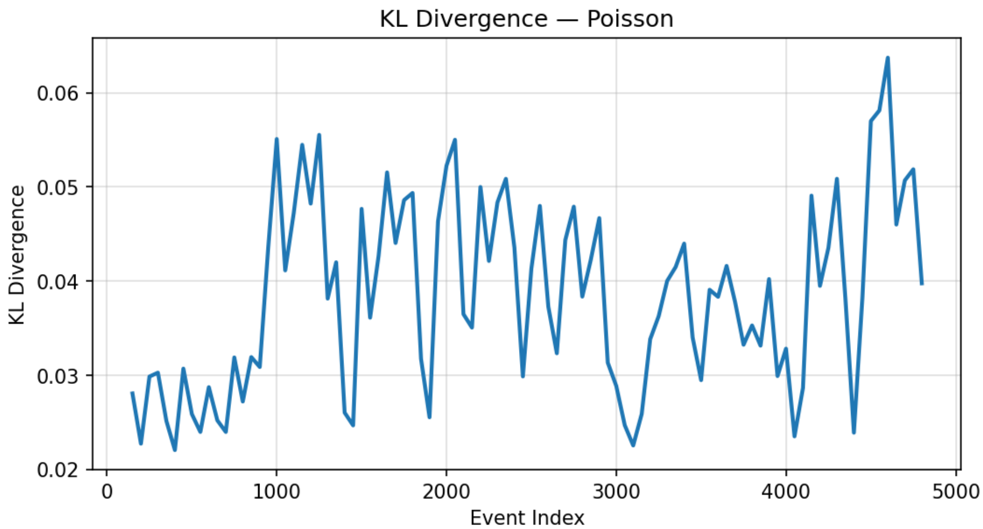

Figure 9: KL divergence for the Poisson baseline. Small, noisy values are consistent with sampling variation around a correct exponential model.

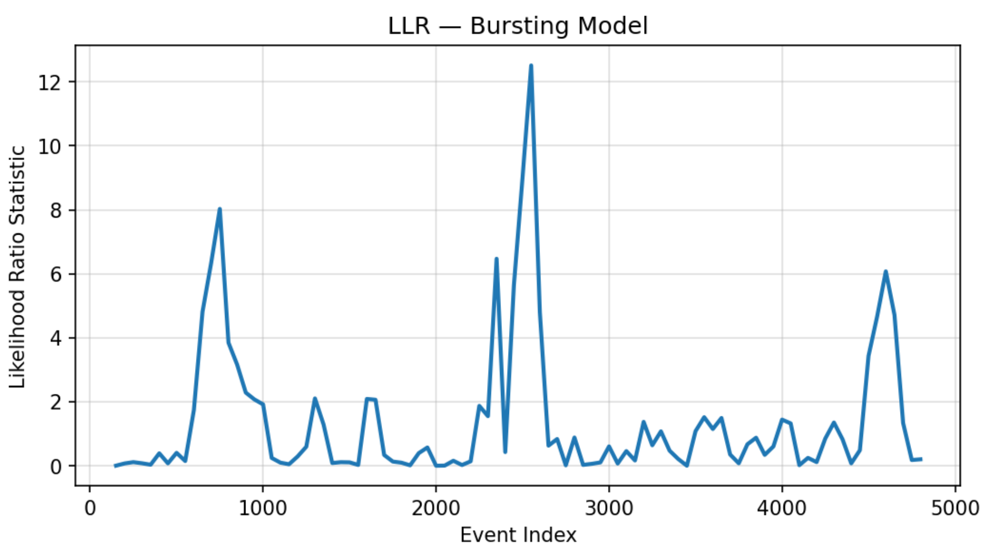

Figure 10: LLR for the bursting regime. Occasional spikes indicate that a Gamma model can sometimes fit a particular window slightly better, but there is no consistent preference.

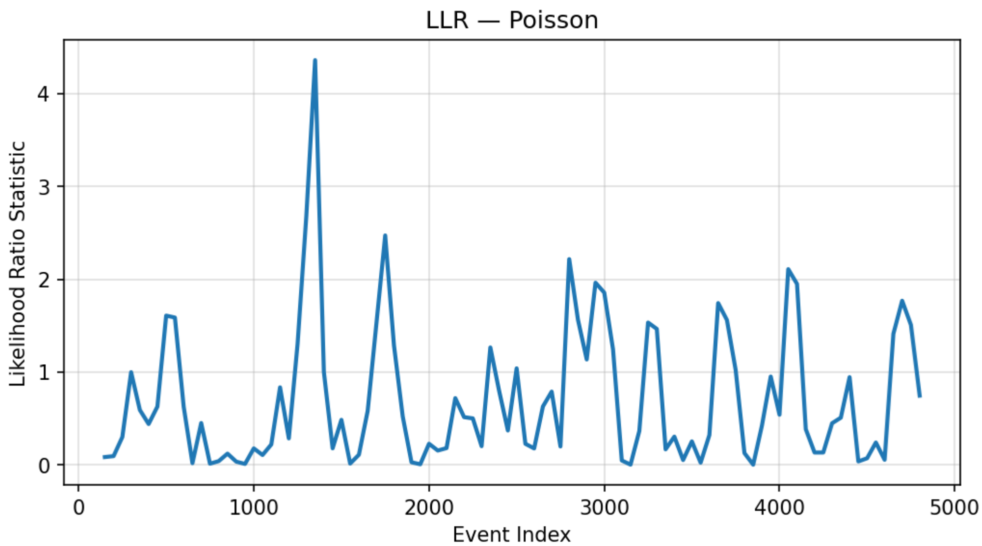

Figure 11: LLR for the Poisson baseline. Values fluctuate near zero, as expected when the data match the exponential model.

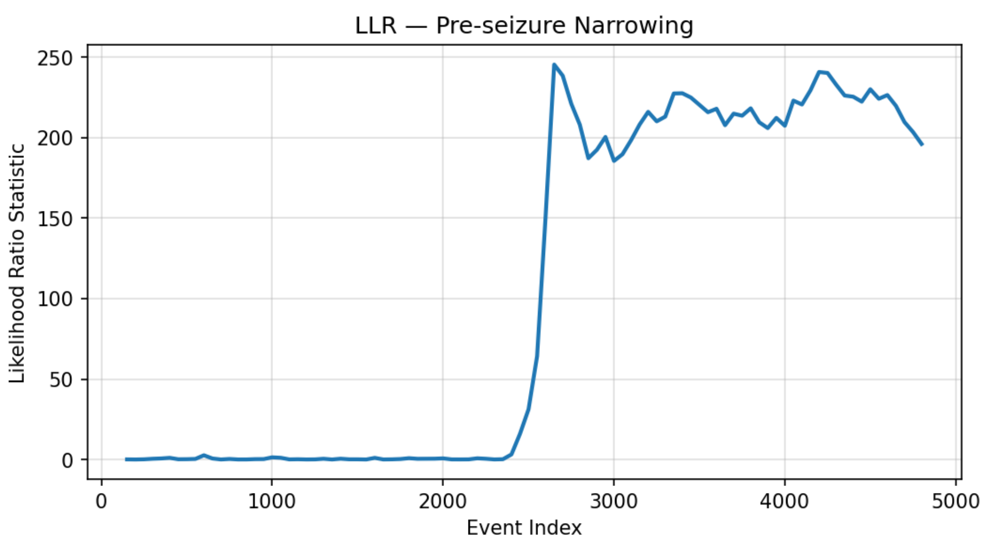

Figure 12: LLR for the pre-seizure narrowing regime. Once narrowing begins, the statistic becomes large and positive, reflecting strong evidence in favor of a less-random Gamma model.